

Size	Method	Live		Non-Live		Multi-Turn		Average	
		Acc.	Format	Acc.	Format	Acc.	Format	Acc.	Format
1.5B	Base	9.33	0.37	12.06	0.12	0.38	0.21	7.25	0.23
	GRPO	63.26 $\pm$ 0.21	99.78 $\pm$ 0.13	77.39 $\pm$ 0.75	99.88 $\pm$ 0.09	2.38 $\pm$ 0.22	24.48 $\pm$ 1.00	47.68 $\pm$ 0.16	74.71 $\pm$ 0.28
	GDPO	68.21 $\pm$ 0.37	99.96 $\pm$ 0.05	79.84 $\pm$ 0.43	100.00 $\pm$ 0.00	3.44 $\pm$ 0.62	64.28 $\pm$ 16.72	50.50 $\pm$ 0.47	88.08 $\pm$ 5.59
	DVAO	68.10	99.85	79.54	100.00	2.25	57.04	49.96	85.63
	GD ² PO-Hard	67.14	100.00	79.81	100.00	5.12	70.28	50.69	90.09
	DARA	68.37 $\pm$ 0.82	99.98 $\pm$ 0.04	79.84 $\pm$ 0.22	100.00 $\pm$ 0.00	4.62 $\pm$ 0.12	88.19 $\pm$ 3.48	50.94 $\pm$ 0.32	96.06 $\pm$ 1.17

Qwen2.5-1.5B-Instruct · BFCL V4 · training step 60 · mean $\pm$ sample SD (%)
 GRPO / DARA n=3, GDPO n=2, DVAO / GD²PO-Hard n=1. Base: untrained reference.
 Retained checkpoints: step-60 RLLA validation Format $\geq$ 80%. Single-seed results: point estimates.
 Average = mean over Live, Non-Live, and Multi-Turn.